What is CNI?

WAS COVERED ALREADY IN K8S NETWORKING CLASS. (REVISION)

CNI = **Container Network Interface**

Simple definition:

CNI gives networking to pods.

Real-Life Analogy

Imagine a new apartment building.

Builder created rooms.

But rooms have:

❌ No electricity

❌ No water

❌ No internet

People cannot communicate.

Kubernetes creates Pods.

But Pods also need:

- IP Address
- Routing
- Communication

Who provides that?

CNI.

Without CNI

Pod gets stuck.

    kubectl get pods

Output:

    ContainerCreating ☹️

Because pod got created but network is missing.

What Does CNI Do?

1. Assign IP

Pod A → 10.244.1.5
Pod B → 10.244.1.6

2. Connect Pods

Pod A
↓
Pod B

Communication becomes possible.

3. Configure Routes

Node1 Pod
↓
Node2 Pod

Cross-node communication.

4. Enforce Network Policies

Can block traffic.

Example:

Frontend
↓
Backend

Allowed.

Random Pod
↓
Backend

Blocked.

POPULAR CNI

1. Flannel

Most Simple CNI

```
Goal:
Just make pods communicate
```

Features:

✅ Pod-to-Pod Networking

❌ Network Policies

❌ Advanced Security

❌ eBPF

Real World

Startup with:

```
10 Nodes
50 Pods
```

No security requirements.

Example:

```
Internal HR Portal
Internal Inventory Tool
```

Flannel is enough.

Why Companies Avoid Flannel Later?

Security team asks:

```
Frontend should not talk to Database
```

Flannel says:

```
Sorry, I don't enforce NetworkPolicy.
```

Then company migrates.

2. Calico

Most Popular Enterprise CNI

Think:

Networking + Security

Features:

✅ Pod Networking

✅ Network Policies

✅ IP Management**Real World Flipkart Example**

Suppose Flipkart has:

Payment Service
Order Service
User Service
Inventory Service

Security team says:

Payment Service
can talk to Database

Inventory Service
cannot talk to Database

Calico enforces this.

✅ BGP Routing

✅ Large Scale Clusters

Why Most Companies Use Calico

Because it gives:

Networking
+
Security

in one solution.

If you enter any enterprise:

TCS
Infosys
Wipro
Accenture
Capgemini

90% chance you'll see Calico.

Even we are using it in our setup

3. Cilium

Newest and Fastest

Built on:

eBPF

instead of traditional iptables.

Real World Analogy

Traditional Networking:

Traffic
↓
Traffic Police
↓
Decision

Cilium:

Traffic
↓
Smart AI Camera
↓
Immediate Decision

Much faster.

Features:

✅ Very Fast

✅ Network Policies

✅ Service Mesh Features

✅ Observability

✅ Security

✅ eBPF

Real World Companies

Common in:

- Uber
- Netflix
- Adobe
- Large SaaS companies

where performance matters.